V5 joint model: independent test evaluation

Run 12877415 · epoch 18 · frozen checkpoint

178 CT scans / 178 patients · 540 nodules · 1423 reader annotations
No training-patient overlap. Ground truth is the official LIDC reader rating.
At objectness ≥ 0.5: 360/540 nodules matched (66.7%); 176 predictions unmatched.

Malignancy classification

Evaluation	Label rule	N	AUC	Accuracy
Reader: official features	Exclude 3	933	0.831	0.776
Reader: official features	Rating ≥ 3	1423	0.674	0.586
Reader: official features	Rating > 3	1423	0.807	0.776
Nodule: matched	Exclude 3	304	0.913	0.868
Nodule: matched	Rating ≥ 3	360	0.835	0.761
Nodule: matched	Rating > 3	360	0.895	0.861
Nodule: all, misses = 0	Exclude 3	423	0.811	0.856
Nodule: all, misses = 0	Rating ≥ 3	540	0.644	0.689
Nodule: all, misses = 0	Rating > 3	540	0.810	0.869
CT: any positive reader	Exclude 3	128	0.912	0.734
CT: any positive reader	Rating ≥ 3	178	0.835	0.601
CT: any positive reader	Rating > 3	178	0.819	0.736
CT: nodule consensus	Exclude 3	137	0.876	0.818
CT: nodule consensus	Rating ≥ 3	178	0.790	0.708
CT: nodule consensus	Rating > 3	178	0.844	0.781

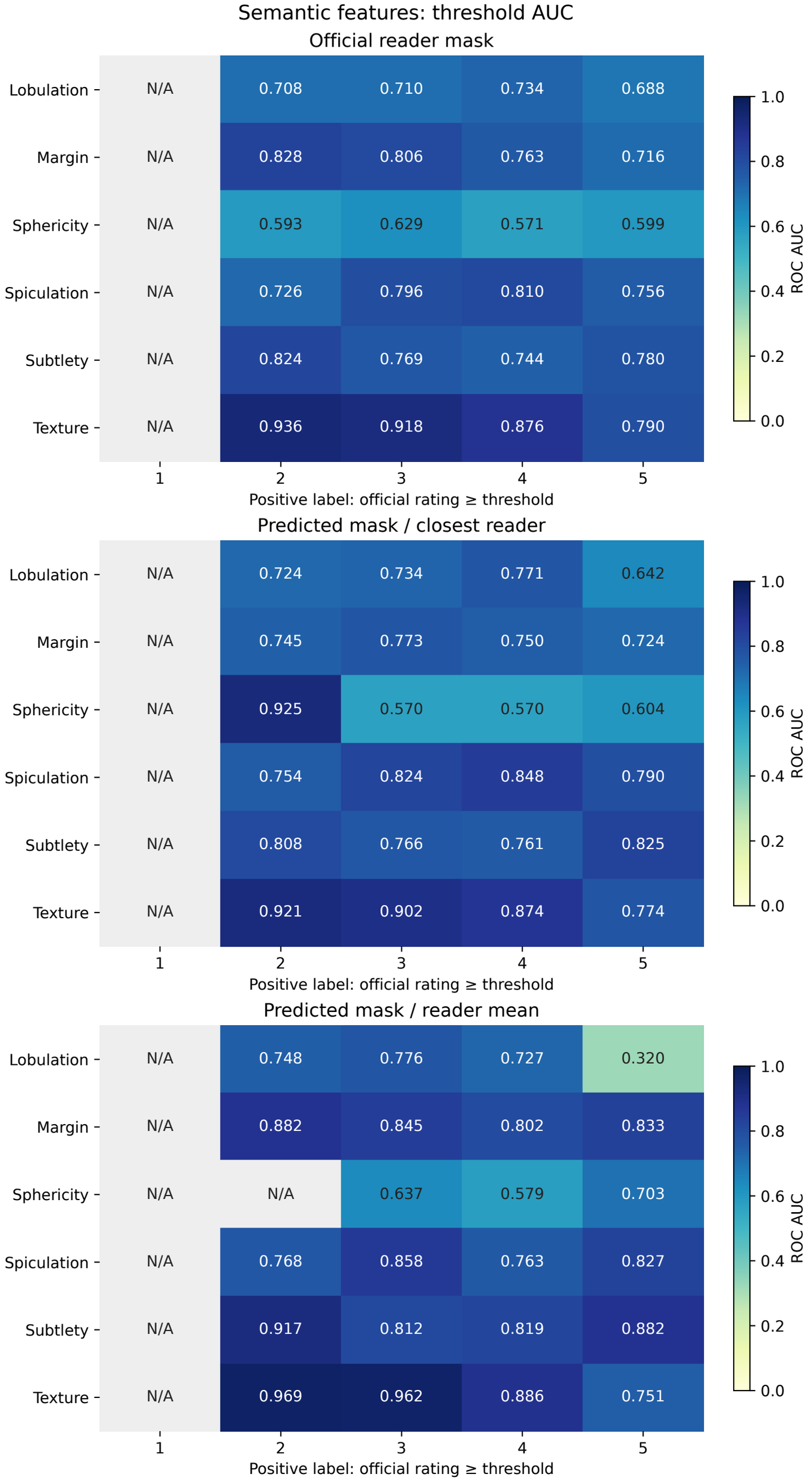
Unmatched prediction exclusions

176 predictions excluded from paired nodule metrics; 21 have predicted malignancy ≥ 0.5.
33 scans have an excluded prediction tied for highest nodule risk.
These predictions remain in the deployed whole-CT risk and count as FROC false positives.

Removing predicted nodules ≤ 5 mm removes 52 predictions,
including 0 classified malignant at probability ≥ 0.5. Additional malignant
detections lost: 1 under exclude-3 / >3; 2 under ≥3.

Among GT nodules ≤ 10 mm (exclude-3), 0/25 malignant nodules
are classified positive, including detection misses in the denominator.

Binary predictions use a fixed probability threshold of 0.5.
The three label rules apply to reader scores or mean nodule scores, as appropriate.
All results are test-set estimates; no thresholds or weights were fitted on this set.


Semantic features: threshold AUC
Official reader mask
Lobulation
Margin
Sphericity
Spiculation
Subtlety
Texture
N/A 0.708 0.710 0.734 0.688
N/A 0.828 0.806 0.763 0.716
N/A 0.593 0.629 0.571 0.599
N/A 0.726 0.796 0.810 0.756
N/A 0.824 0.769 0.744 0.780
N/A 0.936 0.918 0.876 0.790
1 2 3 4 5
Positive label: official rating ≥ threshold
ROC AUC
1.0 0.8 0.6 0.4 0.2 0.0
Predicted mask / closest reader
Lobulation
Margin
Sphericity
Spiculation
Subtlety
Texture
N/A 0.724 0.734 0.771 0.642
N/A 0.745 0.773 0.750 0.724
N/A 0.925 0.570 0.570 0.604
N/A 0.754 0.824 0.848 0.790
N/A 0.808 0.766 0.761 0.825
N/A 0.921 0.902 0.874 0.774
1 2 3 4 5
Positive label: official rating ≥ threshold
ROC AUC
1.0 0.8 0.6 0.4 0.2 0.0
Predicted mask / reader mean
Lobulation
Margin
Sphericity
Spiculation
Subtlety
Texture
N/A 0.748 0.776 0.727 0.320
N/A 0.882 0.845 0.802 0.833
N/A N/A 0.637 0.579 0.703
N/A 0.768 0.858 0.763 0.827
N/A 0.917 0.812 0.819 0.882
N/A 0.969 0.962 0.886 0.751
1 2 3 4 5
Positive label: official rating ≥ threshold
ROC AUC
1.0 0.8 0.6 0.4 0.2 0.0

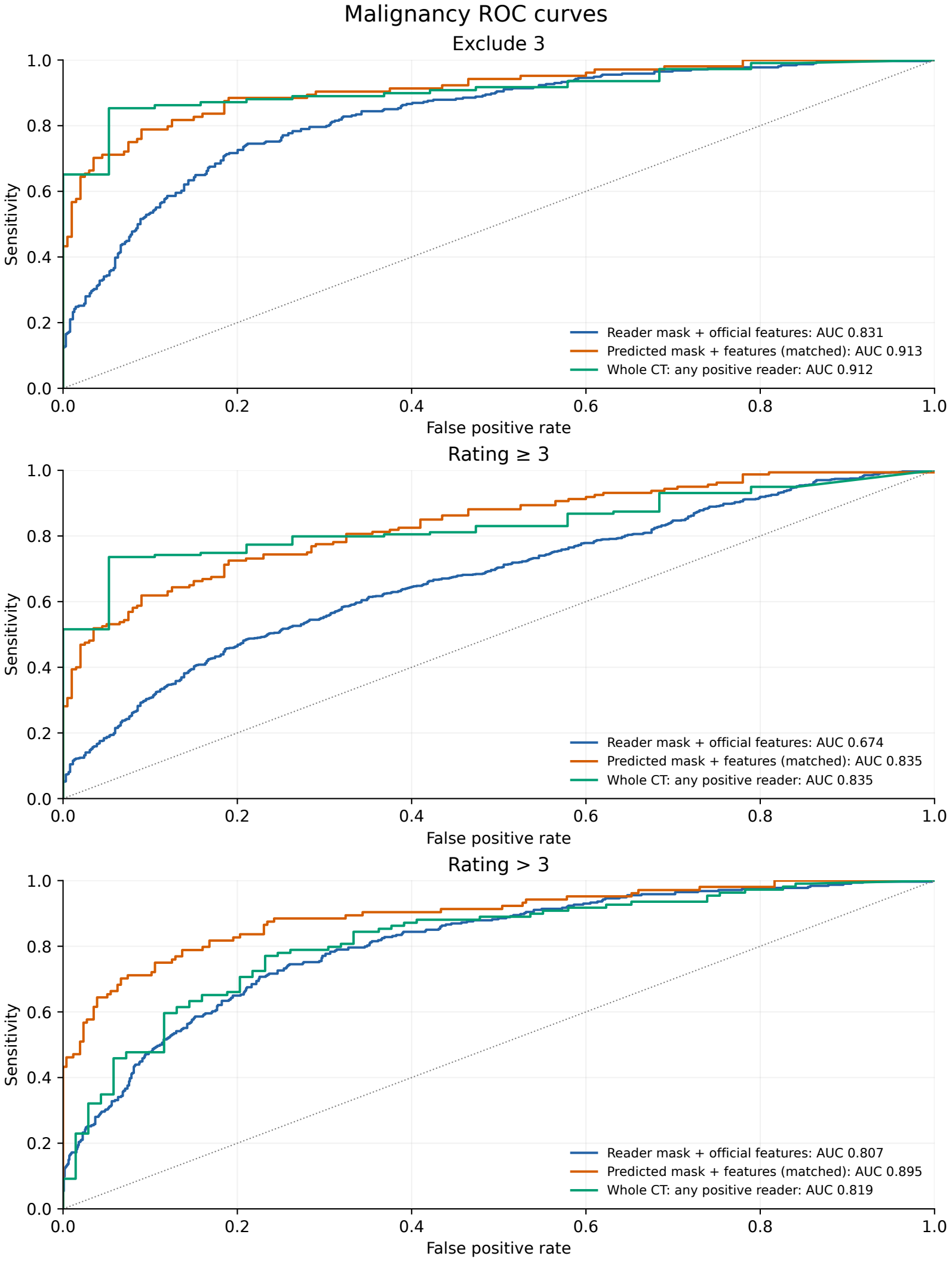
Semantic feature accuracy and rating error

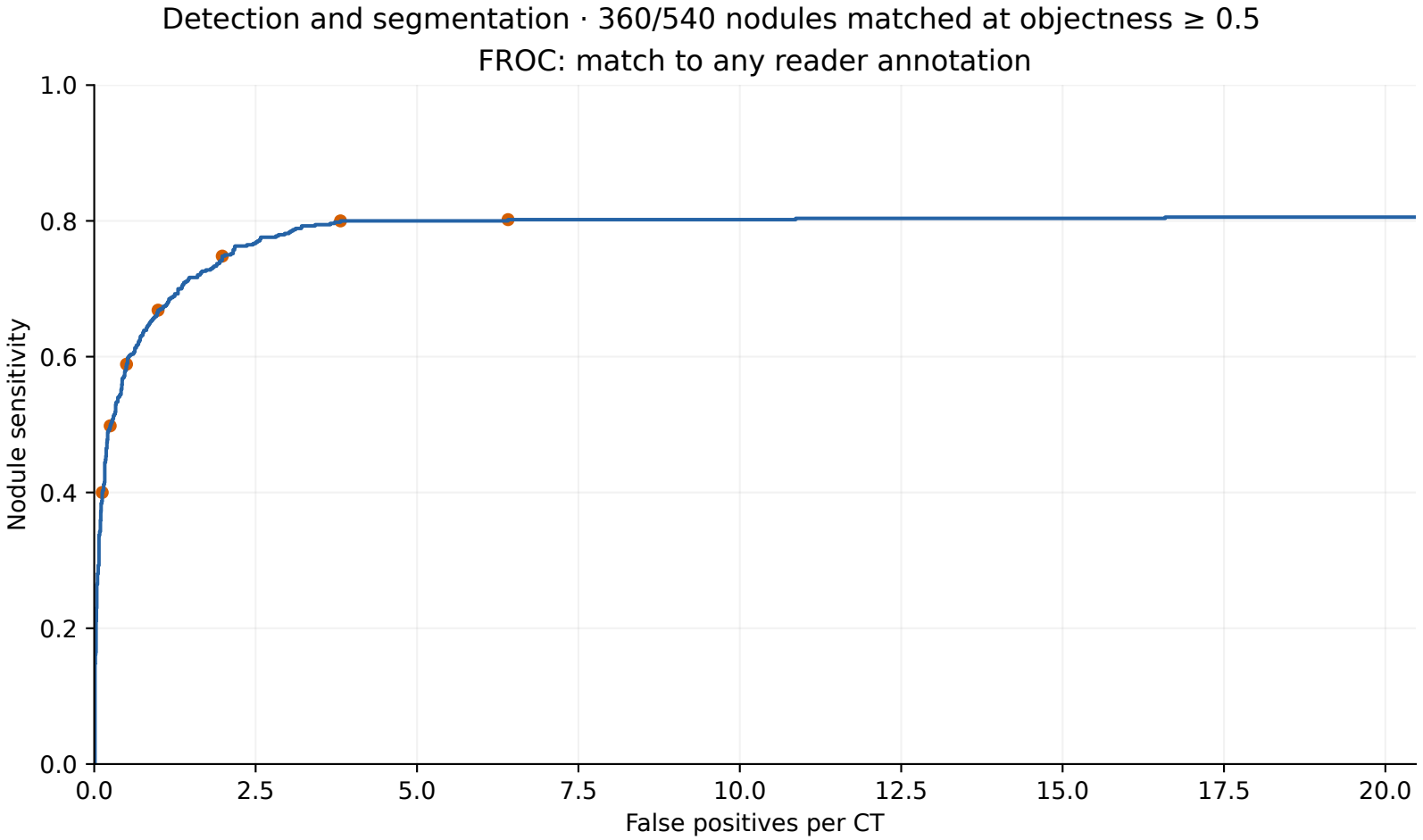
Feature	Reader accuracy	Closest accuracy	Consensus accuracy	Reader MAE	Closest MAE	Consensus MAE
Lobulation	0.645	0.633	0.539	0.772	0.732	0.545
Margin	0.476	0.483	0.519	0.819	0.773	0.562
Sphericity	0.370	0.350	0.447	0.801	0.785	0.566
Spiculation	0.728	0.722	0.631	0.627	0.578	0.446
Subtlety	0.433	0.450	0.511	0.818	0.739	0.529
Texture	0.777	0.794	0.797	0.603	0.529	0.387

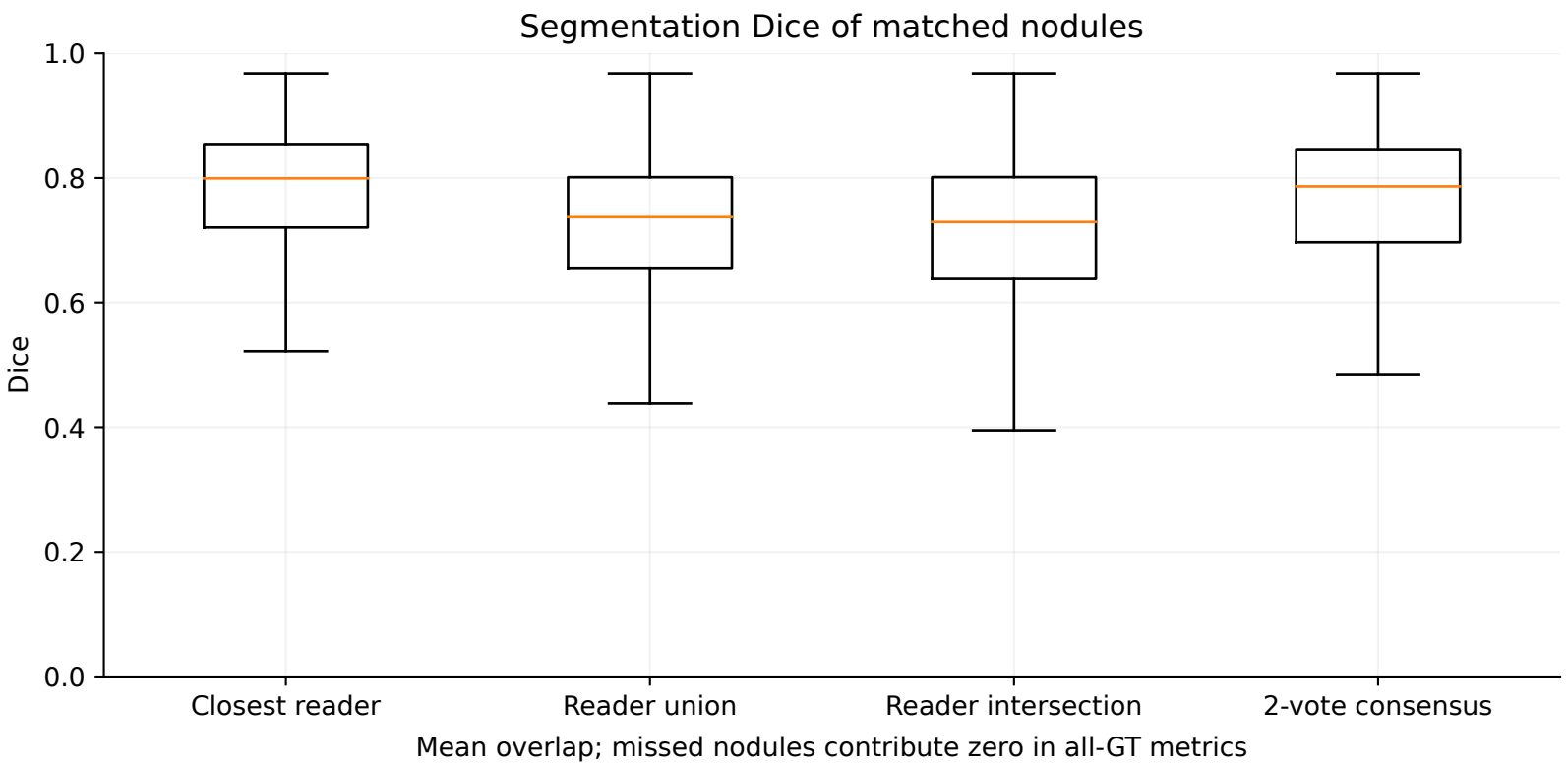
Reader masks: 1423 annotations; predicted masks: 360 matched nodules.

Reader: individual official mask → that reader's score.
Closest: predicted mask → highest-IoU reader's score.
Consensus: predicted mask → mean of all reader scores.

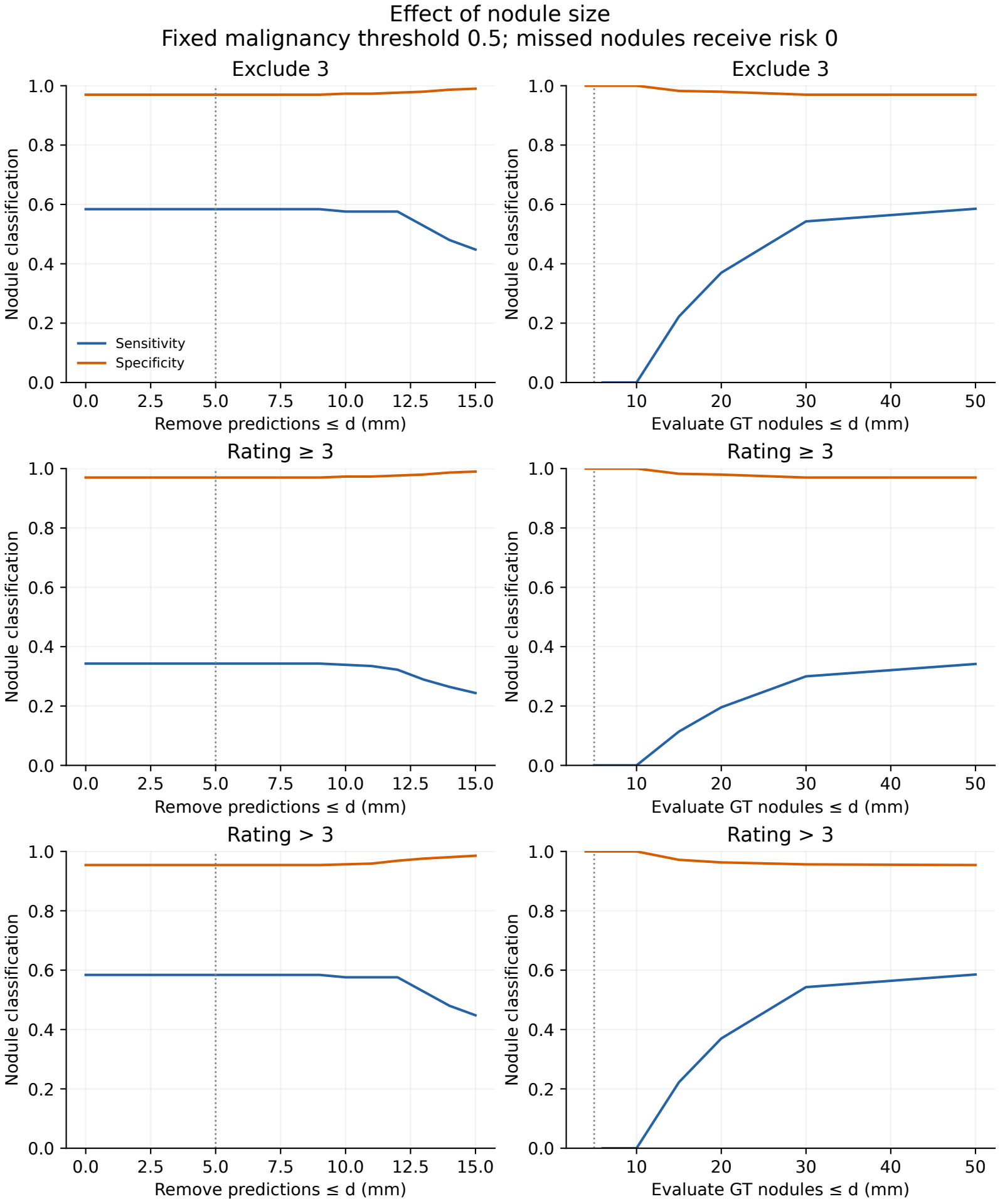
Accuracy compares the most probable class with the target class (1–5).
Consensus targets are rounded half-up for accuracy; MAE uses unrounded scores.
Missed nodules have no semantic prediction and are counted separately in detection.


Malignancy ROC curves
Exclude 3
Sensitivity
False positive rate
Reader mask + official features: AUC 0.831
Predicted mask + features (matched): AUC 0.913
Whole CT: any positive reader: AUC 0.912
Rating ≥ 3
Sensitivity
False positive rate
Reader mask + official features: AUC 0.674
Predicted mask + features (matched): AUC 0.835
Whole CT: any positive reader: AUC 0.835
Rating > 3
Sensitivity
False positive rate
Reader mask + official features: AUC 0.807
Predicted mask + features (matched): AUC 0.895
Whole CT: any positive reader: AUC 0.819





Reference	Matched Dice	Matched IoU	All-GT Dice	All-GT IoU
Closest	0.773	0.642	0.515	0.428
Union	0.708	0.563	0.472	0.375
Intersection	0.711	0.565	0.474	0.377
Consensus	0.759	0.625	0.506	0.417


Effect of nodule size
Fixed malignancy threshold 0.5; missed nodules receive risk 0
Exclude 3
Exclude 3
Rating ≥ 3
Rating ≥ 3
Rating > 3
Rating > 3
Nodule classification
Sensitivity
Specificity
Remove predictions ≤ d (mm)
Evaluate GT nodules ≤ d (mm)
0.0
0.2
0.4
0.6
0.8
1.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
10
20
30
40
50

Evaluation definitions

Cohort and labels

Every testing CT and physical nodule is included, including mean malignancy = 3.
The risk bank and neural weights are frozen; no test patient fitted the bank.
Training excluded nodules with mean malignancy = 3; the two inclusion rules also evaluate these nodules in the test cohort.
Malignancy is derived from official reader ratings, as requested; no biopsy labels are available in the supplied table. These are not pathology-confirmed outcomes.

Per-annotation component evaluation

Each reader mask is independently applied to CT. The semantic network receives masked CT and mask. The malignancy stage receives that same reader's six official features plus the radiomics baseline computed from the reader's mask.
An additional table reports risk from predicted semantic features on reader masks.

Per-nodule evaluation

Whole-CT inference receives no GT boxes, masks or ratings. Predictions match physical nodules one-to-one at mask IoU ≥ 0.1 against any reader. Matching first maximizes valid pair count, then total IoU. The closest reader has the largest IoU; ties use annotation ID. Union, intersection and 2-vote masks use all readers.
Unmatched predictions are excluded from paired semantic/malignancy metrics.
Missed GT remains in detection and all-nodule metrics; its operational risk is 0.
Consensus semantic labels use reader means, rounded half-up for exact accuracy.
Threshold labels use the unrounded mean; t = 1 AUC is undefined.
Sparse endpoints: closest-reader sphericity at t = 2 has one negative; consensus lobulation at t = 5 has one positive. Counts are in the threshold CSV.

Per-volume evaluation and FROC

Default whole-CT risk retains unmatched predictions and uses the model's object-weighted noisy-OR within each bank member, followed by ensemble averaging.
CT metrics include both any positive reader and any positive mean nodule score.
When excluding 3, a scan with an indeterminate label and no positive is unknown.
ROC plots use any positive reader; the tables also include oracle matched-only risk.
FROC sweeps all post-NMS candidates from object threshold 0; duplicates count as false positives. All 540 physical nodules remain in the sensitivity denominator.

Size and interpretation

Size is the maximum physical 3D Feret diameter of occupied voxel cells. GT size averages reader diameters. The left size plots remove predicted masks ≤ d and rematch; the right plots evaluate GT nodules whose size is ≤ d without filtering.
All size analyses keep missed malignant nodules in the denominator.